

Promise

Easy and simple use
BP lowering is good
Cholesterol lowering is satisfactory
Blood thinning is effective
Tolerability is good

Challenge

Is it safe in the long run?
Will it deny flexibility to the physician?
Will it encourage people to smoke and eat unhealthy food?
Will it overmedicate?
Will it reduce mortality?

Pharmaceuticals Ltd, which sponsored the study, was part of the steering committee that designed the trial but had no role in data collection, analysis, interpretation or writing.

After 12 weeks, the results showed that each of the components of the polypill had done what it was intended to do: the statin had reduced cholesterol, three BP-lowering drugs had done their job and aspirin had brought down blood clotting. The key upshots were: the polypill could cut CVD risk by half in healthy people with a single risk factor; that side-effects were minimal; that the different drugs did not adversely affect each others' actions; and that the polypill could treat all the CVD risk factors in a patient. "A smoker without a history of high BP or high cholesterol, as

a doctor tell which component caused the side-effect? In that case, can the polypill be adapted to individualised therapy? How safe is it in the long run? Would it reduce mortality? Will the availability of a single magic pill encourage people to continue to smoke, eat unhealthy and indulge in other lifestyle excesses? Will it medicalise human life, with anyone over the age of 55 tossing down a pill every day? "The polypill holds tremendous promise," says Xavier, "But lifestyle modification is the key to reduce CVD and should be actively promoted. In some situations drugs are necessary and there the polypill can be a very useful option."

Cadila's 'Polycap' is ready to hit the markets "in a matter of weeks". The researchers are firming up plans

HOW THE POLYPILL WORKS

Some Risk Factors	Drugs Used to Treat	Cuts Heart Risk By	Stroke Risk By
LDL cholesterol	Simvastatin (20 mg) to lower cholesterol	27%	8.1%
Diastolic BP	Atenolol, betablocker Ramipril, ACE inhibitor Hydrochlorothiazide, diuretic (all in half dose) to treat high BP	24%	33%
Platelet stickiness	Aspirin (100 mg) to thin the blood	32%	16%
COMBINED risk reduction...		62%	48%

Between March 2007 and August 2008, the trial assessed the effects of nine different pills in various drug combinations on 2,053 patients at 50 centres across the country. The patient profile was that of a group that has been rising rapidly in the past decade with the upsurge of obesity—middle-aged (45-80 years) men and women without previous record of CVD, but with at least one risk factor (high BP, obesity, high cholesterol, diabetes or smoking). A primary concern was to ensure that the size of the pill should be manageable. It came to be called 'Polycap' because the final pill took the shape of a capsule. Cadila

presently defined, was nonetheless treated for both and was able to stomach the treatment," adds Pais. The polypill is also likely to win in cost-effectiveness. "Most of the components are available in a generic form. And the polypill is likely to be much more affordable than buying the five constituents separately," says Xavier.

That's not the end of the story. The journey of the polypill has just started and the road ahead is full of challenges. Will people adhere to a polypill better than they would to five different pills? What if patients face life-threatening side-effects for drugs they actually don't need? How would



“Our drug trial is an important first step and has vast implications for global health.”

P. PAIS & D. XAVIER,
St. John's Medical
College, Bangalore

to go ahead with more robust studies to fine-tune issues of efficacy and safety. Meantime, one can only hope that policies and laws against tobacco will get tighter; fast-food joints and food manufacturers will cut down on trans fats; the 300-million-strong Indian middle class will trade in their quick-fix consumption habits for a healthy lifestyle; fewer people will spend their workdays behind a desk; in the kitchen, convenience foods will take a backseat; and urban sprawl will make room for safe and pleasurable physical activities in everyday life. If not, at least, the polypill will come in handy. ■